

1. Introduction to PC Architecture

PC Architecture kya hai?

PC (Personal Computer) Architecture ka matlab hai computer ke andar lage different hardware components ki **design, structure aur organization**, aur ye components ek-dusre ke saath kaise kaam karte hain.

Simple Definition

PC Architecture = Computer ke parts ka arrangement aur unke beech communication kaise hota hai.

Jab hum computer par koi kaam karte hain, jaise:

- MS Word open karna
- Game khelna
- Internet browse karna

toh computer ke different hardware components milkar kaam karte hain. Isi complete working ko **PC Architecture** kaha jaata hai.

Main Components of Computer

1. CPU – Central Processing Unit

CPU ko computer ka **Brain** kaha jaata hai.

CPU ke main kaam:

- Instructions ko execute karna
- Calculations karna
- Decisions lena
- Hardware devices ko control karna

Example

Agar hum calculator mein **20 + 30** karte hain:

Keyboard → CPU → Monitor

- Keyboard input deta hai.
 - CPU calculation karta hai.
 - Monitor result **50** display karta hai.
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2. Memory

Memory ka use **data aur instructions store** karne ke liye hota hai.

Mainly memory ke do types:

A. RAM – Random Access Memory

RAM ek **temporary memory** hai.

Features:

- Data temporarily store hota hai.
- Computer OFF hone par RAM ka data erase ho jaata hai.
- RAM bahut fast hoti hai.

Example

Jab hum **MS Word** open karte hain, toh MS Word RAM mein load hota hai.

Computer OFF karne par RAM ka temporary data remove ho jaata hai.

Easy Trick:

RAM = Temporary Memory

B. ROM – Read Only Memory

ROM ek **permanent/non-volatile memory** hoti hai.

Features:

- Important permanent instructions store karti hai.
- Power OFF hone ke baad bhi data normally retain rehta hai.

Easy Trick:

ROM = Permanent Memory

3. Input Devices

Input devices ka use computer ko **data ya instructions dene** ke liye hota hai.

Examples:

- Keyboard
- Mouse
- Scanner
- Microphone

Input = Computer ko data dena

4. Output Devices

Output devices computer ke processed result ko **user tak pahunchate/display karte hain**.

Examples:

- Monitor
- Printer
- Speaker
- Projector

Output = Computer se result lena

5. Storage Devices

Storage devices ka use data ko **long-term/permanently store** karne ke liye hota hai.

Examples:

- Hard Disk
- SSD
- Pen Drive
- DVD

Working of Computer

Basic working:

User → Input Device → CPU → Memory → Output Device

Example:

Hum keyboard se data enter karte hain → CPU usko process karta hai → Memory data/instructions ko handle karti hai → Output device result show karta hai.

2. Introduction to Intel Pentium

Intel Pentium Intel company ke dwara developed microprocessors ki ek family hai.

Pentium ko **1993** mein introduce kiya gaya tha. Ye Intel 486 processor ka successor tha.

Us time Pentium processors ko high-performance processors maana जाता था.

Pentium naam kahan se aaya?

"Pentium" word Greek word **Penta** se related hai, jiska meaning **five (5)** hota hai.

Processor kya hai?

Processor ko **CPU (Central Processing Unit)** bhi kaha jaata hai.

Ye computer ka **brain** hota hai.

Processor ke main kaam:

- Instructions process karna
- Calculations karna
- Programs run karna
- Hardware aur software ke operations ko control karna

Example

Jab hum **Google Chrome** par click karte hain:

Click → CPU instruction receive karta hai → Process karta hai → Program memory mein load hota hai → Screen par Chrome open hota hai.

Without processor, computer properly kaam nahi kar sakta.

3. Introduction to PC Hardware

Hardware kya hai?

Computer ke woh physical parts jinhe hum **dekh aur touch** kar sakte hain, unhe **Hardware** kehte hain.

Simple Definition

PC Hardware = Computer system ke saare physical components.

Examples:

- Monitor
- Keyboard
- Mouse
- CPU/Processor
- Motherboard
- RAM
- Hard Disk

- SSD
 - Printer
 - Speakers
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Types of PC Hardware

PC hardware ko mainly 4 categories mein divide kar sakte hain:

1. **Input Devices**
 2. **Output Devices**
 3. **Processing Devices**
 4. **Storage Devices**
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Main Hardware Components

1. Motherboard

Motherboard computer ka **main circuit board** hota hai.

Computer ke almost saare important hardware components motherboard se connected hote hain.

Functions:

- CPU ko connect karta hai.
- RAM ko connect karta hai.
- Storage devices ko connect karta hai.
- Other hardware components ko connect karta hai.

Easy Example

Motherboard ko computer ki **backbone** samajh sakte hain.

2. CPU / Processor

CPU computer ka **brain** hota hai.

Ye instructions ko process karta hai aur computer ke operations ko control karta hai.

Functions:

- Data process karna
- Calculations karna
- Programs run karna

- Hardware ko control karna

Example

MS Word open karne par CPU required instructions ko process karta hai.

3. RAM

RAM ek **temporary memory** hai.

Isme currently running programs aur required data temporarily store hota hai.

Computer OFF hone par RAM ka temporary data lost ho jaata hai.

Example

Agar aap kisi document par kaam kar rahe hain aur use save nahi kiya, toh computer suddenly OFF hone par unsaved work lost ho sakta hai.

4. ROM

ROM ek **non-volatile memory** hai.

Isme computer ke startup/important instructions stored ho sakte hain.

Power OFF hone ke baad bhi ROM ka data normally retain rehta hai.

5. Hard Disk (HDD)

HDD ek storage device hai jo data ko long-term store karta hai.

Isme hum store kar sakte hain:

- Documents
 - Photos
 - Videos
 - Software
 - Other files
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4. What is a Segment?

Segment kya hai?

Segment memory ka ek particular section/block hota hai, jisme specific type ka data ya program store kiya jaata hai.

Simple Definition

Segment = Memory ka ek section jiska specific purpose hota hai.

Easy Example

School library mein different sections ho sakte hain:

- Science books → ek section
- Mathematics books → ek section
- Story books → ek section

Isi tarah memory ko bhi different sections/segments mein organize kiya ja sakta hai.

Segmentation ka use kyun hota hai?

Segmentation:

- Memory ko organize karti hai.
- Programs ko manage karna easy banati hai.

- Large programs ko smaller sections mein divide karne mein help karti hai.
 - Data aur program ke different parts ko manage karne mein help karti hai.
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5. Introduction to Addressing

Addressing kya hai?

Addressing wo method hai jisse CPU memory mein stored data ya instruction ki location ko identify karta hai.

Computer ki har memory location ka ek **unique address** hota hai.

Easy Example

Jaise city mein har house ka unique house number hota hai:

- House No. 101
- House No. 102
- House No. 103

Waise hi memory mein har location ka apna address hota hai.

Address kya hai?

Address = Memory location ka identification number/location.

CPU address ki help se:

- Data read karta hai.
- Data write/store karta hai.
- Instructions ko execute karta hai.

Addressing ki need

Addressing CPU ko help karti hai:

- Correct data find karne mein.
 - Instructions read karne mein.
 - Results store karne mein.
 - Correct memory location access karne mein.
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6. Introduction to Register

Register kya hai?

Register CPU ke andar present **bahut small aur very fast storage location** hota hai.

Registers temporary basis par:

- Data
- Instructions
- Memory addresses

store karte hain.

Simple Definition

Register = CPU ke andar ki very fast temporary memory.

Registers RAM se faster hote hain kyunki ye directly CPU ke andar hote hain.

Registers ka use kyun hota hai?

Registers:

- Data temporarily store karte hain.
- Instructions ko hold karte hain.

- Memory addresses store karte hain.
- CPU processing ko fast banate hain.
- Computer ki performance improve karne mein help karte hain.

Example: 20 + 30

CPU calculation ke time:

1. **20** ek register mein load hota hai.
2. **30** doosre register mein load hota hai.
3. ALU dono numbers ko add karta hai.
4. Result **50** register mein store hota hai.
5. Zarurat ke according result memory mein store kiya ja sakta hai.

Easy Trick

Register = CPU ki working table

Jaise hum kaam karte time required things table par rakhte hain, waise CPU

processing ke time required data registers mein rakhta hai.

7. Introduction to Assembly Language

Computer directly **Machine Language** ko understand karta hai.

Machine language mein instructions **0 aur 1 (binary)** ke form mein hoti hain.

Humans ke liye binary mein programming karna difficult hota hai.

Isliye Assembly Language use ki gayi.

Assembly Language kya hai?

Assembly Language ek **low-level programming language** hai jo machine language ke bahut close hoti hai.

Isme binary ki jagah short words/codes use kiye jaate hain, jinhe **Mnemonics** kaha jaata hai.

Examples of Mnemonics:

- MOV

- ADD
- SUB

Simple Definition

Assembly Language = Machine language ke close language jo mnemonic instructions use karti hai.

Assembler kya hai?

Assembler ek program hai jo Assembly Language instructions ko **Machine Language (machine code)** mein convert karta hai.

Flow:

Assembly Language → Assembler → Machine Code

Features of Assembly Language

- Mnemonics use karti hai.
- Machine language se comparatively easy hai.

- Hardware ke close hoti hai.
 - Low-level language hai.
 - Assembler ki need hoti hai.
 - Hardware-level programming mein useful hai.
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Machine Addressing

Machine Addressing kya hai?

Machine addressing wo method hai jisse CPU memory mein **data ya instructions ki location identify** karta hai.

Har memory location ka unique address hota hai.

CPU in addresses ki help se:

- Data read karta hai.
- Data store karta hai.
- Instructions execute karta hai.
- Results ko correct location par store karta hai.

Simple Example

Jaise kisi ghar ka address batata hai ki ghar exactly kahan hai, waise hi memory address CPU ko batata hai ki required data **memory mein kahan stored hai.**

Debugging

Program likhte time errors/bugs aa sakte hain.

In errors ko find karke correct karne ke process ko **Debugging** kehte hain.

Simple Definition

Debugging = Program ke errors ko find aur fix karna.

Bug kya hai?

Program mein koi error ya mistake ho, toh use **Bug** kehte hain.

Example

Agar program ka expected result:

10 + 20 = 30

hai, lekin program:

10 + 20 = 40

de raha hai, toh program mein bug/error hai.

1. Data Definition

Assembly Language mein kisi variable ko use karne se pehle uske liye memory define/reserve karni padti hai.

Is process ko **Data Definition** kaha jaata hai.

Data definition assembler ko batati hai:

- Variable ka naam kya hai.
- Data ka type kya hai.
- Kitni memory required hai.
- Kya value store karni hai.

Example

NUM DB 10

Yahan:

- **NUM** = Variable name
- **DB** = Define Byte
- **10** = Value

Matlab **NUM** naam ke variable mein **10** value ko one byte ke form mein store karna.

2. Directives

Directive kya hai?

Directive ek special command/instruction hai jo **assembler ko batata hai ki program ko kaise prepare/handle karna hai.**

Directive normally CPU ke liye executable machine instruction nahi hoti.

Simple Definition

Directive = Assembler ko diya gaya command.

Directives ka use

- Variables define karne ke liye
- Memory reserve karne ke liye

- Constants define karne ke liye
- Program ke sections define karne mein
- Assembler ko necessary information dene ke liye

Important Difference

Instruction → CPU execute karta hai.

Directive → Assembler process karta hai.

Addressing Format

Addressing Format kya hai?

Addressing format batata hai ki instruction mein required data **kahan se milega** ya uski location kaise specify ki gayi hai.

Simple Definition

Addressing Format = Instruction mein data ki location specify karne ka tarika.

COM Program

COM Program kya hai?

COM Program ek simple executable program format hai jo mainly **DOS operating system** mein use hota tha.

COM file ki extension:

.COM

hoti hai.

Simple Definition

COM Program = .COM extension wali simple Assembly executable program file.

COM Program ki Features

- File extension **.COM** hoti hai.
- Simple program format hai.
- Code, data aur stack same memory segment ke context mein hote hain.
- Maximum size approximately **64 KB** hota hai.
- DOS systems mein commonly used tha.

- Simple programs ke liye useful tha.

★ Quick Revision

Topic	Easy Meaning
PC Architecture	Computer parts ka structure aur working
CPU	Computer ka brain
RAM	Temporary memory
ROM	Non-volatile/permanent memory
Input Device	Computer ko data dene wala device
Output Device	Result dene wala device
Storage	Data ko long-term store karna
Motherboard	Main circuit board
Segment	Memory ka specific section

Topic	Easy Meaning
Address	Memory location ki identification
Register	CPU ke andar very fast temporary storage
Assembly Language	Low-level language using mnemonics
Assembler	Assembly ko machine code mein convert karta hai
Debugging	Errors ko find aur fix karna
Bug	Program ki error
Data Definition	Variable aur memory define karna
Directive	Assembler ko diya command
Addressing Format	Data ki location specify karne ka method

Topic	Easy Meaning
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COM Program	DOS ka simple .COM executable program
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★ **Important Exam Points**

1. CPU = Brain of Computer
2. RAM = Temporary/Volatile Memory
3. ROM = Non-volatile Memory
4. Register = CPU ke andar very fast memory
5. Assembly Language = Low-level language
6. Assembler = Assembly Language
→ Machine Code
7. Debugging = Errors find + fix
8. Bug = Program error
9. DB = Define Byte
10. COM file extension = .COM
11. COM program maximum size = 64 KB

12. Address = Memory location ko identify karta hai
13. Motherboard = Main circuit board
14. Input = Data computer ko dena
15. Output = Processed result receive karna